Topic D: historical and current proof dependencies

Human Compression project

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This packet records the dependencies of the accepted Topic D edition. Its three source contracts concern commuting involutions, Suzuki regular polyhedra, and the Hall–Janko geometry census. The graph preserves all 37 source records and all 19 bound statement/proof contexts. It does not assign proof weights, calculate compression factors, or add a source result.

An arrow labelled *proof use*, *definition use*, *hypothesis use*, *proof assembly*, or *retained input use* points from a prerequisite to the argument that uses it. The other arrow types record attribution, comparison, diagnostics, formal evidence or execution. They are excluded from logical closures. Historical arrows document the dependencies asserted or used in the source; they do not certify a flawed intermediate argument retrospectively.

The three current routes

Commuting involutions (L066). The orbit data of a marked Klein four-group gives five multiplicities, faithfulness and parity. Reversible coordinates yield the ordered counts. Endpoint exchange is then counted with its fixed-orbit correction. Ordinary symmetric/alternating automorphism theorems remain retained inputs. Degree six is handled by full automorphism orbits, middle-involution stabilizers, generation and every intersection test. The complete exceptional program is a necessary premise of this finite classification. The regular-polytope correspondence is retained when interpreting the two symmetric-group survivors and the absence of an alternating-group survivor.

Suzuki polyhedra (L074). Classical finite Suzuki root groups, ovoids, automorphisms, canonical subfields and maximal subgroups remain visible premises. The application proves the coordinate normalization, positions a containing subfield, identifies the generated group, checks the intersection property and proves the rank restriction. These local steps identify actual field support; the common support theorem does not supply them for free. Möbius inversion and Frobenius orbit lengths then give the uniform count. The odd-orbit lemma justifies dividing by two for duality. The stabilizer-fiber theorem also gives the nested formula printed in the source. Finite field and $q = 8$ diagnostics are not premises of these all-parameter proofs.

Hall–Janko geometries (L104). The complete subgroup catalogue, exact group operations and full automorphism group remain retained. Coset realization, the typed Helly criterion and arbitrary-Borel panel recognition connect subgroup tuples to the exact geometric axioms. Transporters and stabilizer fibers make maximal-residue augmentation complete. Four required programs perform augmentation, full automorphism/type fusion and separate final geometry verification using the common predicates. The resulting census is $8, 19, 3$ in ranks two, three and four; the empty terminal level excludes higher ranks. Catalogue completeness is not inferred from a successful search.

What is shared, and what remains local

The common extension theorem classifies data above one retained base orbit by the action of its stabilizer. It is used for exceptional commuting triples, Suzuki first-coordinate fibers, and the augmentation search. The combined support theorem adds invariant admissibility, an equivariant support poset and finite fixed-point counts. Its applications still owe their admissibility, generation, support and freeness checks.

The finite-action centralizer calculation and the panel-recognition arguments serve different branches. The full faithful-action normalizer exact sequence is proved in the common paper, but the selected application routes do not require that full statement. It remains present in the complete paper; the graph does not silently delete its written cost. No splitting of that sequence is asserted.

Historical qualifications

The L066 endpoint-reversal statement is recovered with a repaired proof: conjugating one endpoint alone does not establish simultaneous conjugation of the pair. The source's self-duality remark is retained only at the endpoint level. Its final alternating-group exclusion remains an exclusion of *regular* polyhedra. Earlier centralizer and polyhedron results keep their original attribution.

L074 explicitly quotes earlier Suzuki and Lüneburg results. The current paper retains their specified finite structure. It does not rederive the full arbitrary-field Bruhat statement or the general ovoid plane-section assertion. The printed torus-order coprimality lemma is unused by the new intersection proof. Fermat's congruence is classical background. The old almost-simple rank theorem is an input to the historical proof but is rederived in the current application; this earns no separate source credit.

The L104 principal census does not include complete recovery of the printed rank-four diagrams, all automorphism/correlation groups, the doubling relation or every supplement annotation. Those assertions have historical nodes and explicit excluded routes. The old Magma/geometry programs are not identified with the present GAP implementation or TomLib catalogue. Their complete construction and software costs remain unknown.

Programs, Lean and the review boundary

The graph retains five complete required mathematical programs, three complete diagnostics, and two complete execution drivers. Diagnostics have validation arrows; drivers have operational arrows. The driver's historical M22 branch is preserved in the actual file but is not a Topic D theorem route. No program is shortened for this graph.

The original two Lean modules supply ten scoped reports: eight concern fiber actions and odd-period arguments, and two give an abstract rank obstruction. The later support supplement supplies 28 reports, comprising the unchanged first twelve and sixteen further reports. The first twelve are not added again. These are evidence for precisely scoped common statements, not 38 independent mathematical results or formal proofs of every application. Source-specific generation, the finite-field support adapter, the number-theoretic divisor bridge and catalogue completeness are not certified by that supplement.

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